Question ID: 9d4270fe

Question

A company that creates and sells tape dispensers calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. The equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. Which statement is the best interpretation of $2.00x$ in this context?

Answer

A. The monthly sales revenue, in dollars, from selling x tape dispensers

B. The monthly sales revenue, in dollars, from each tape dispenser sold

C. The monthly cost, in dollars, of creating each tape dispenser

D. The monthly cost, in dollars, of creating x tape dispensers

Question ID: 7a5a74a6

Question

$3(2x-6)-11=4(x-3)+6$

If x is the solution to the equation above, what is the value of $x-3$?

Answer

A. $\frac{23}{2}$

B. $\frac{17}{2}$

C. $\frac{15}{2}$

D. $-\frac{15}{2}$

Question ID: aa85b138

Question

$2n+6=14$

A tree had a height of 6 feet when it was planted. The equation above can be used to find how many years n it took the tree to reach a height of 14 feet. Which of the following is the best interpretation of the number 2 in this context?

Answer

A. The number of years it took the tree to double its height

B. The average number of feet that the tree grew per year

C. The height, in feet, of the tree when the tree was 1 year old

D. The average number of years it takes similar trees to grow 14 feet

Question ID: 15daa8d6

Question

$$2x + 16 = a(x + 8)$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

Question ID: 12ee1edc

Question

$(b-2)x=8$

In the given equation, b is a constant. If the equation has no solution, what is the value of b ?

Answer

A. 2

B. 4

C. 6

D. 10

Question ID: 018a2704

Question

If $46 = 16 + 2(x - 8)$, what is the value of $2(x - 8)$?

Answer

A. 16

B. 23

C. 30

D. 38

Question ID: 70e29454

Question

$a(3-x)-b=-1-2x$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?

Answer

A. $a=2$ and $b=1$

B. $a=2$ and $b=7$

C. $a=-2$ and $b=5$

D. $a=-2$ and $b=-5$

Question ID: f09097b1

Question

An agricultural scientist studying the growth of corn plants recorded the height of a corn plant at the beginning of a study and the height of the plant each day for the next 12 days. The scientist found that the height of the plant increased by an average of 1.20 centimeters per day for the 12 days. If the height of the plant on the last day of the study was 36.8 centimeters, what was the height, in centimeters, of the corn plant at the beginning of the study?

Question ID: 620abf36

Question

If $5(x+4)=4(x+4)+29$, what is the value of $x+4$?

Answer

A. -4

B. 25

C. 29

D. 33

Question ID: 4f669597

Question

$$2(p+1)+8(p-1)=5p$$

What value of p is the solution of the equation above?

Question ID: ce314070

Question

If $4x - \frac{1}{2} = -5$, what is the value of $8x - 1$?

Answer

A. 2

B. $-\frac{9}{8}$

C. $-\frac{5}{2}$

D. -10

Question ID: feb78194

Question

A museum rents tablets to visitors. The museum earns revenue of $\$14$ for each tablet rented for the day. On Wednesday, the museum earned $\$406$ in profit from renting tablets after paying daily expenses of $\$112$. How many tablets did the museum rent on Wednesday?
$(\text{profit} = \text{total revenue} - \text{total expenses})$

Question ID: ce6b52d8

Question

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

Question ID: 76f29fa5

Question

The cost to rent a commercial fishing boat from a certain company is $\$950$ for the first 2 hours and an additional $\$50$ per hour for each hour after the first 2 hours. If the total cost to rent the commercial fishing boat from the company for t hours, where $t > 2$, is $\$1{,}100$, which equation represents this situation?

Answer

A. $950(t-2)+50t=1{,}100$

B. $950(2t)+50t=1{,}100$

C. $950+50(t-2)=1{,}100$

D. $950+50(2t)=1{,}100$

Question ID: 36ab4122

Question

Megan’s regular wage at her job is p dollars per hour for the first 8 hours of work in a day plus 1.5 times her regular hourly wage for work in excess of 8 hours that day. On a given day, Megan worked for 10 hours, and her total earnings for that day were $137.50. What is Megan’s regular hourly wage?

Answer

A. $11.75

B. $12.50

C. $13.25

D. $13.75

Question ID: 5ad9eff0

Question

The width of a rectangular dance floor is w feet. The length of the floor is 6 feet longer than its width. Which of the following expresses the perimeter, in feet, of the dance floor in terms of w ?

Answer

A. $2w+6$

B. $4w+12$

C. w^2+6

D. w^2+6w

Question ID: 45bba652

Question

If $2(x-5)+3(x-5)=10$, what is the value of $x-5$?

Answer

A. 2

B. 5

C. 7

D. 12

Question ID: 8c515062

Question

A candle is made of 17 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 6 ounces of wax remain in this candle, for how many hours has it been burning?

Answer

A. 3

B. 6

C. 24

D. 44